

Name Key

## Physics 390 In-Class Activity 14

### MRI SIGNAL AMPLITUDE

$$\text{Signal Amplitude}(x, y) \propto M_{xy}(T_E) \propto \rho(x, y)(1 - e^{-TR/T_1})e^{-TE/T_2}$$

| Field strength (T) | Tissue         | $T_1$ (ms) | $T_2$ (ms) | Proton density |
|--------------------|----------------|------------|------------|----------------|
| 3.0                | White matter   | 1080       | 70         | 0.61           |
|                    | Gray matter    | 1820       | 100        | 0.69           |
|                    | Arterial blood | 1932       | 275        | 0.72           |
|                    | CSF            | 3817       | 1442       | 1.0            |

The equation above describes how spin echo amplitude depends on experimental parameters ( $TR$  and  $TE$ ) and tissue parameters ( $T_1$  and  $T_2$ ). Assume, for a given experiment, that  $TR$  and  $TE$  are 2000 ms and 70 ms, respectively.

(A) Calculate the spin echo amplitude for arterial blood and white matter at 3.0 T, using values from the table above. Write down the numerical values of all three terms in the amplitude (not just the final answer), so you can see how the terms contribute!

$$SA_{\text{blood}} \propto 0.72 \times (1 - e^{-2000/1932}) \times (e^{-70/275})$$

$$SA_{\text{blood}} \propto 0.72 \times 0.64 \times 0.78 = 0.36$$

$$SA_{\text{wm}} \propto 0.61 \times (1 - e^{-2000/1080}) \times e^{-70/70}$$

$$SA_{\text{wm}} \propto 0.61 \times 0.84 \times 0.37 = 0.19$$

(B) Which component will be brighter and which will be darker?

Arterial blood has a larger SA and will be brighter  
WM is darker

(C) What kind of weighting is being used? Justify your answer!

$T_2$ -weighting

The last ( $T_2$ ) term  $e^{-TR/T_2}$  is more than 2-fold different (ratio-wise). The other terms differ less.